

Robust classification of pleural effusion on ChestMNIST under uncertain labels

Phase III: experimental protocol, results and discussion

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Abstract

This document is the Phase III contribution to the TP4 article: the experimental protocol, the measured results and their discussion. Seven classical classifiers are trained on 10 000 ChestMNIST radiographs and evaluated on the untouched official validation and test splits for the detection of pleural effusion, a finding present in 12.3 percent of the test images. The three families with the highest validation average precision are then retrained under a simulated annotation error, five percent of the positive training labels flipped to negative, each in a standard and in a more strongly regularised configuration, over three sampling seeds. The sections below report what was measured, under which protocol, and what the measurements do and do not support.

1 Experimental protocol

1.1 Task, data and splits

The experiments use ChestMNIST, the chest radiograph subset of MedMNIST v2 [9], which repackages the NIH ChestX-ray14 collection [8] as 28×28 grayscale thumbnails with 14 binary thorax findings per image. The findings are not mutually exclusive: a training image carries 0.72 positive findings on average and up to nine at once. A 14-class decision is therefore not defined on this data, and the task retained here is the binary detection of pleural effusion, read from column 2 of the label matrix.

Table 1 reports the partition. The official MedMNIST train, validation and test splits are preserved exactly, so that the numbers of this article remain comparable with the published MedMNIST baselines. Only the training split is subsampled, to 10 000 images, by stratified sampling with seed 42; the subsample therefore keeps the 11.80 percent effusion prevalence of the full training split, and the retained row indices are written to `results/train_indices_seed42.npy`. The cap is a budget decision: seven model families, a 2×2 robustness design and three seeds have to fit in the session, and the kernel machine is quadratic in the number of training examples.

Two preprocessing facts are worth recording because every later number depends on them. First, the arrays are dense `uint8` with no missing-value sentinel; the absence of missing pixels and missing labels is verified programmatically rather than assumed, and is reported in `results/dataset_description.json`. Sec-

Table 1: ChestMNIST partition for the effusion target. The splits are those distributed by MedMNIST v2; the training subsample of 10 000 images used below is drawn from the train row with stratification.

Split	Images	Effusion positive	Negative	Prevalence
Train	78 468	9 261	69 207	0.1180
Val	11 219	1 292	9 927	0.1152
Test	22 433	2 754	19 679	0.1228

ond, each image is flattened to 784 intensities rescaled to $[0, 1]$ and then standardized per feature. The scaler, and the 50 component principal basis used in the representation check of section 2.3, are fitted on the training subsample only, so no statistic of the validation or test images enters the representation.

1.2 Models and hyperparameter budget

Seven families are evaluated: logistic regression trained by stochastic gradient ascent on the penalised conditional log-likelihood, Gaussian naive Bayes, k -nearest neighbours, a CART decision tree, histogram gradient boosting [4], a soft margin linear support vector machine and an RBF kernel support vector machine [1]. All of them are fitted on the same 784 dimensional standardized features, so the comparison measures the inductive bias of each family and not a difference of representation. Implementations come from scikit-learn 1.8.0 [5], except the stochastic gradient ascent learner, which is written for this work because the lab sheet asks for the update to be implemented explicitly.

The hyperparameter budget is capped at three candidates per family, as agreed in the common protocol: $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}\}$ for the L2 strength of the gradient ascent learner, $K \in \{3, 5, 11\}$ for the neighbourhood size, a maximum depth in $\{3, 5, 10\}$ for the tree, and $C \in \{0.1, 1, 10\}$ for both support vector machines. Naive Bayes and boosting are run in one fixed configuration. The budget is small on purpose: a wider search would make the comparison of section 2.1 a comparison of search effort rather than of model families.

Six of the seven families expose a posterior through `predict_proba`. The two support vector machines expose only the signed distance to the separating hyperplane. Ranking metrics are computed from the posterior where it exists and from the signed margin otherwise. This is legitimate because ROC AUC and average precision depend on the induced ordering of the examples alone, not on the scale of the score, so the two

families remain comparable without fitting a Platt calibration on top of the margins. It does mean, however, that the margin based models are read at threshold 0 and the probabilistic ones at 0.5, which section 2.2 returns to.

1.3 Metrics and selection rule

Every run is scored with accuracy, precision, recall, F1, ROC AUC and average precision, computed by a single evaluation layer (`src/evaluate.py`) so that seven families, four experimental conditions and three seeds cannot drift apart through differing conventions. Matthews correlation and specificity are recorded in the result files as well.

All model selection in Phase III uses *validation average precision*. The reason is the prevalence. With 2754 positive and 19679 negative test images, a classifier that always answers “no effusion” already reaches 0.877 accuracy, so accuracy cannot separate a useful detector from a constant one. The ROC curve is almost as insensitive, for a reason that is arithmetic rather than conceptual: its false positive rate is divided by the 19679 negatives, so one hundred additional false positives move it by 0.005, whereas at a recall of 0.5 the same hundred errors move precision from 0.500 to 0.483. The precision-recall curve divides by the quantity a radiologist cares about, the number of positive calls made, and its no-skill level is the prevalence itself rather than a fixed diagonal. This is the standard argument for preferring precision-recall to ROC under class imbalance [2, 6], and average precision is its scalar summary. The test split is read once per configuration, after the choice has been made on validation.

Because the default cut-off of each family is calibrated for balanced data, F1 is also reported at a second operating point, the threshold that maximises F1 on the validation split and is then transferred unchanged to the test split. The two columns separate two questions that a single F1 conflates: how well a model orders patients, and where its decision boundary happens to sit.

1.4 Simulation of uncertain labels

Task 3.2 asks for a controlled annotation error. In the training subsample the minority class is the effusion positive class, with 1180 of 10000 images, so a five percent corruption rate flips 59 labels from positive to negative and lowers the training prevalence from 0.1180 to 0.1121. The flip is deliberately asymmetric, from positive to negative, for two reasons. It matches the dominant failure mode of report mined annotation, in which a small effusion is simply not mentioned and the image is recorded as negative, and it is the harder direction for a minority class, since it removes evidence from the class that is already scarce.

Three constraints make the resulting comparison interpretable and are enforced in code rather than by convention. The corrupted row indices are drawn once per seed and shared by every model, so a per-

Table 2: The 2×2 design of Task 3.2, repeated for each of the three selected families and each of the three seeds. The upper left cell reproduces the baseline of table 3.

Training labels	Standard	Robust
Unaltered	fit, evaluate	fit, evaluate
5% of the minority class flipped	fit, evaluate	fit, evaluate

formance gap between two families can never be an artefact of two different corruptions; the indices are saved to `results/flipped_indices_seed*.npy`. Validation and test labels are never altered, so every metric is measured against the reference annotation. And the corruption is recorded, not inferred: the number of flipped rows and the prevalence before and after are logged with each run.

The three families with the highest validation average precision are retrained in the four cells of table 2. Crossing the label condition with the capacity of the hypothesis class, rather than reporting the corrupted and regularised cell alone, is what makes the result readable: the clean and robust cell measures the price paid for extra regularisation when nothing is wrong with the labels, the corrupted and standard cell measures the damage, and only the difference of differences supports a claim about robustness rather than about a configuration that happens to be better everywhere.

The robust configurations constrain the hypothesis class more tightly than the standard ones, in the direction indicated by the lab sheet: a stronger L2 penalty for the gradient ascent learner, $\lambda \in \{10^{-2}, 10^{-1}, 1\}$; a smaller margin penalty for the support vector machines, $C \in \{10^{-2}, 10^{-3}\}$; shallower trees with larger leaves for the tree based models; a wider neighbourhood, $K \in \{25, 51, 101\}$, for k -nearest neighbours.

Which robust candidate is retained is decided once, on the validation average precision of the clean primary-seed run, and is then held fixed for the corrupted condition and for the two replication seeds. Re-selecting it under corruption would describe what a practitioner actually does, but it would make the difference between the two cells of a row mix a change of labels with a change of configuration, and the question asked here is whether one fixed, more constrained model degrades less than one fixed, less constrained one. Freezing the configuration also means the standard and the robust rows of ?? are read the same way. These configurations are experimental hypotheses, and ?? reports whether the data supports them.

1.5 Replication, tracking and reproducibility

The robustness experiment is repeated with seeds 42, 7 and 2024. The seed jointly controls the training subsample, the corrupted indices and the internal randomness of the estimators, so the spread across seeds bounds the sampling variability of the whole pipeline and not of one of its stages. The baseline sweep of sec-

tion 2.1 and the choice of the robust hyperparameters are performed on seed 42 only.

Every run is logged to an MLflow experiment [10] backed by a local SQLite store (Task 3.3). Each record carries the model family, all its hyperparameters, the seed, the feature space, the training subset size, the corruption rate and the number of flipped labels; the eight metrics on validation and on test; the fit and prediction wall clock times; the experimental cell as a queryable tag; and, as artifacts, the serialised estimator, the list of flipped indices and the trace of the candidates examined during selection. The run history is reproducible from the two scripts, so the database itself is not committed. Results are additionally appended to `results/runs_robustness.jsonl` as they complete, which makes an interrupted sweep resumable and keeps the tables reproducible without MLflow. Library versions are pinned in `requirements.txt`; the tree based models in particular break ties differently across scikit-learn minor releases.

2 Results and discussion

2.1 The seven baselines

Table 3 reports the retained candidate of each family, and figs. 1 and 2 show the corresponding curves on the test split. Histogram gradient boosting leads on validation average precision with 0.2719, ahead of the RBF kernel machine at 0.2417 and the linear support vector machine at 0.2158. The k -nearest neighbours rule with $K = 11$ follows closely at 0.2128, then the depth 5 tree at 0.2058, Gaussian naive Bayes at 0.1845 and the gradient ascent logistic model at 0.1833. The first three were therefore carried into Task 3.2.

The informative comparison is not the raw value but the lift over the no-skill level, because average precision inherits the prevalence of the split it is measured on, and the two held out splits do not have the same prevalence (0.1152 on validation, 0.1228 on test). On test, boosting reaches 2.24 times the no-skill level and the RBF machine 2.25 times, while the gradient ascent logistic model reaches 1.65. Two conclusions follow. The models do extract a real signal, between 1.6 and 2.3 times what chance gives on this prevalence. And that signal is modest: a precision near 0.26 at a recall near 0.5 (table 4) means three false alarms for every correct call, which is not a usable screening operating point.

Three observations deserve to be separated, because each is a different way the imbalance distorts a familiar metric.

Accuracy ranks a constant classifier first. Six of the seven models sit between 0.8522 and 0.8772 test accuracy, inside the band around the 0.8772 that an always-negative answer scores. The highest accuracy of the table, 0.8772, belongs to the RBF kernel machine, and it is exactly that trivial value because at margin 0 the model makes *no* positive call at all on the 22 433 test images: 0 true positives and 0 false

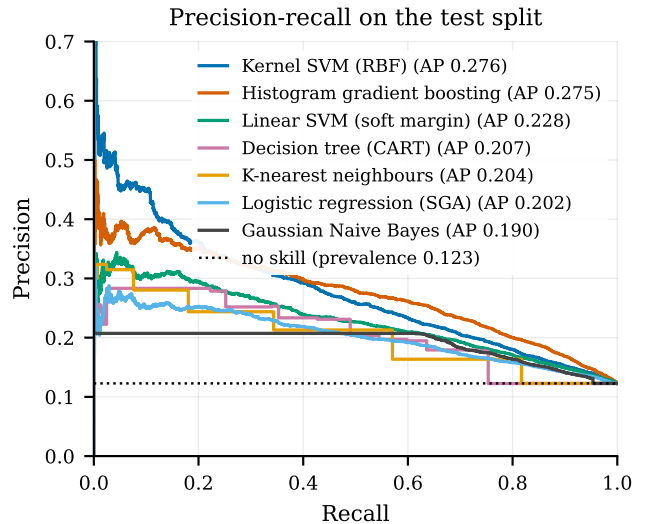


Figure 1: Precision-recall curves on the test split. The dotted line is the no-skill level, equal to the 0.1228 prevalence. The vertical axis is truncated at 0.70: below a recall of about 0.02 the curves are decided by a handful of images and swing between 0 and 1.

positives. The model with the best test average precision of the table is therefore also the one whose hard predictions carry no information. Accuracy appears in table 3 for completeness and is used nowhere else in this work.

F1 at the default cut-off inverts the ranking.

Gaussian naive Bayes has the second worst average precision (0.1898 on test) and the best default F1 (0.2995), while boosting has the best average precision and an F1 of 0.0870. The cause is visible in the number of positive calls: naive Bayes issues 9681 of them on 22 433 images, against 348 for boosting and 0 for the RBF machine. Multiplying 784 near-duplicate likelihood terms over strongly correlated pixels drives the posterior to the ends of $[0, 1]$, which leaves the default 0.5 cut-off in a region where almost every image is called positive; the F1 maximising threshold selected on validation for that model is 1.0000, the boundary of the interval, which is the signature of a saturated posterior. Naive Bayes is not better at detecting effusion, it is differently miscalibrated. Any conclusion drawn from F1 at the default threshold on this task would be an artefact of calibration.

ROC and precision-recall disagree where it matters.

Boosting dominates on ROC AUC (0.7447 against 0.7175 for the RBF machine) but the two are level on test average precision (0.2753 against 0.2763). The reversal is the arithmetic of section 1.3: the 19 679 negative test images in the denominator of the false positive rate absorb precisely the errors that the precision-recall view charges to the model. Figure 2 also compresses the seven models into a narrow band, which would suggest that the choice of family hardly matters, whereas fig. 1 separates boosting and the kernel machine from the rest over most of the recall range. This is the concrete form, on this dataset, of the ar-

Table 3: The seven baselines on the primary seed. Each row is the candidate of its family with the highest validation average precision; the test columns are computed once, after that choice. The score type is *prob.* when the ranking metrics come from `predict_proba` and *margin* when they come from `decision_function`. Accuracy, precision, recall and F1 are taken at each family’s default cut-off, which section 2.2 shows to be the wrong reading of this task.

Model	Configuration	Score type	Val. AP	Test split					
				Acc.	Prec.	Rec.	F1	AUC	AP
Histogram gradient boosting	leaves 31	prob.	0.2719	0.8738	0.3879	0.0490	0.0870	0.7447	0.2753
Kernel SVM (RBF)	$C = 1$	margin	0.2417	0.8772	0.0000	0.0000	0.0000	0.7175	0.2763
Linear SVM (soft margin)	$C = 0.1$	margin	0.2158	0.8725	0.3174	0.0338	0.0610	0.6911	0.2281
K-nearest neighbours	$K = 11$	prob.	0.2128	0.8740	0.3237	0.0243	0.0453	0.6732	0.2041
Decision tree (CART)	depth 5	prob.	0.2058	0.8732	0.2443	0.0156	0.0294	0.6695	0.2075
Gaussian Naive Bayes	$\varepsilon = 10^{-9}$	prob.	0.1845	0.6117	0.1923	0.6761	0.2995	0.6626	0.1898
Logistic regression (SGA)	$\lambda = 10^{-5}$	prob.	0.1833	0.8522	0.2498	0.1017	0.1445	0.6606	0.2024

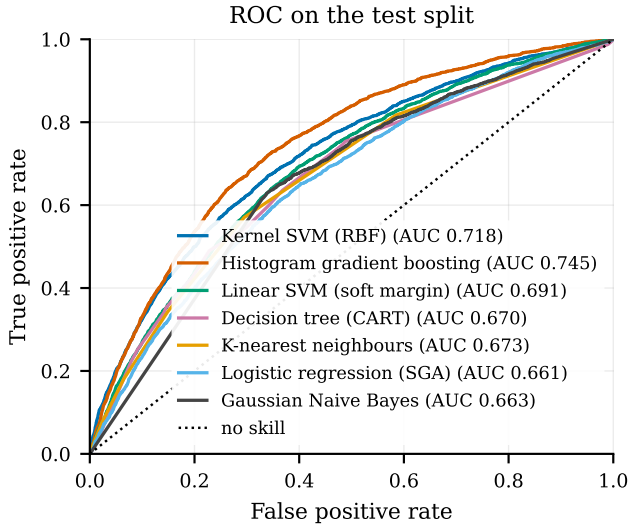


Figure 2: ROC curves for the same seven models and the same score vectors as fig. 1. The two views do not agree on the ordering of the top two models, and the ROC view makes all seven look closer to one another than the precision-recall view does.

gument of Davis and Goadrich [2] and of Saito and Rehmsmeier [6].

One cost belongs next to the scores. The RBF machine needs 146.7s to score the 22 433 test images against 2.1s for boosting, a factor of 69, because its decision function is a sum over several thousand support vectors in 784 dimensions. At equal average precision, boosting is the better engineering choice.

2.2 Where the decision boundary sits

Table 4 reads the same seven models at a second operating point, the threshold that maximises F1 on validation, transferred unchanged to test. Every model improves and the degenerate cases disappear: boosting goes from 0.0870 to 0.3639, the RBF machine from 0.0000 to 0.3411, the linear machine from 0.0610 to 0.3070. Naive Bayes, first at the default cut-off with 0.2995, moves only to 0.3099 and falls back to the middle of the table. The ordering by F1 at the selected threshold then broadly follows the ordering by average precision, which is the expected behaviour: once the threshold stops being an uncontrolled variable, the thresholded metric measures the same under-

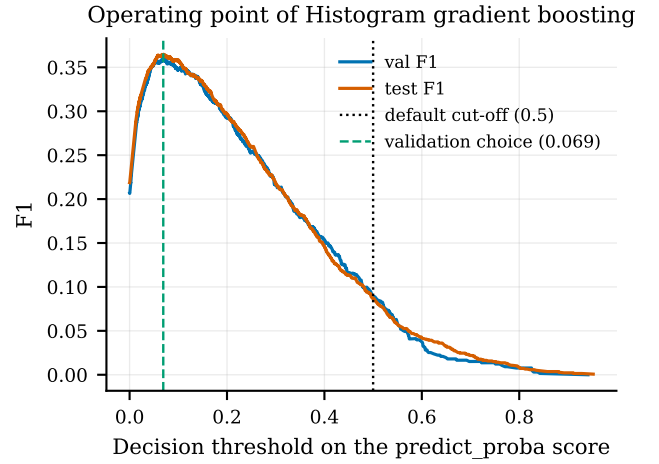


Figure 3: F1 of the boosted model against its decision threshold, on validation and on test. The default 0.5 cut-off lies far to the right of the maximum, which on this prevalence sits near 0.07. The two curves peak at nearly the same place, so the threshold transfers across splits.

lying ranking quality.

Figure 3 shows why. For the boosted model the F1 maximum lies at a posterior of 0.0691, not at 0.5, and the validation and test curves peak at nearly the same place, so the choice transfers. The practical reading is that a thresholded metric on an imbalanced task reports the sum of two things, the quality of the ranking and the accident of where the library put the cut-off, and that the second term dominates here. This is why average precision drives every selection decision in this work, and why ?? reports F1 at the validation selected point rather than at the default one.

2.3 Does reducing the dimension help?

The two families whose behaviour depends most directly on the ambient dimension were rerun on a 50 component principal projection fitted on the training subsample, retaining 95.3 percent of the variance (table 5). Neither improves. The $K = 5$ neighbours rule falls from 0.1822 to 0.1776 validation average precision and the RBF machine from 0.2417 to 0.2344. The expectation behind the ablation, that Euclidean distances and an isotropic kernel would behave better once 784 correlated pixel coordinates are compressed, is not met. The discriminative signal lies partly in

Table 4: Test precision, recall and F1 at the default cut-off of each family (0.5 on a posterior, 0 on a margin) and at the cut-off that maximises validation F1. The test split is never used to choose the threshold. Every model gains, and the two models that made almost no positive call at the default cut-off gain the most.

Model	Default cut-off			Cut-off chosen on validation		
	Prec.	Rec.	F1	Prec.	Rec.	F1
Histogram gradient boosting	0.388	0.049	0.087	0.264	0.587	0.364
Kernel SVM (RBF)	0.000	0.000	0.000	0.264	0.483	0.341
Linear SVM (soft margin)	0.317	0.034	0.061	0.235	0.442	0.307
K-nearest neighbours	0.324	0.024	0.045	0.213	0.570	0.310
Decision tree (CART)	0.244	0.016	0.029	0.231	0.490	0.314
Gaussian Naive Bayes	0.192	0.676	0.299	0.207	0.614	0.310
Logistic regression (SGA)	0.250	0.102	0.145	0.205	0.484	0.288

Table 5: Representation check on the primary seed. The principal basis is fitted on the training subsample only and retains 95.3 percent of the variance.

Model	Pixels (784)		PCA (50)	
	AUC	AP	AUC	AP
KNN, $K = 5$	0.6336	0.1819	0.6317	0.1810
Kernel SVM, $C = 1$	0.7175	0.2763	0.7086	0.2683

directions that principal component analysis discards for carrying little variance, which is plausible on this modality: variance in a chest radiograph is dominated by patient size, framing and exposure, none of which is the costophrenic angle. The projection does buy computation, since the RBF machine then scores the test split in 21.9s instead of 146.7s, a factor of 6.7, for 0.008 of average precision. That trade is worth taking in deployment and not worth taking when the point is to compare model families, so the main tables keep the 784 dimensional features.

2.4 What these experiments do not settle

Four limits bound every claim made above, and they are stated here rather than left for a reader to infer.

The images are thumbnails and the features are pixels. A pleural effusion is read on a radiograph as blunting of the costophrenic angle and a fluid meniscus at the lung base, a local geometric cue occupying a small part of the image. At 28×28 that region covers a few pixels, and a flat vector of 784 intensities gives none of the models any notion of adjacency. The absolute scores reported here should be read as what classical classifiers extract from a thumbnail, not as an estimate of what is achievable on full resolution radiographs. The comparison that this work supports is the one between families, conditions and metrics, all measured under one fixed representation.

Published numbers are not directly comparable. The MedMNIST authors report a convolutional baseline on the same 28×28 ChestMNIST images [9], but averaged over the 14 findings and as area under the ROC curve, whereas this study reports a single

finding and relies on average precision. A difference in target, in metric and in the number of training images is enough to make a direct ranking meaningless. Published figures and the figures of this work are therefore kept in separate tables, and no claim of superiority or inferiority is made across them.

The clean condition is not a clean reference.

The annotations of ChestX-ray14, which ChestMNIST redistributes [8], were extracted from free-text radiology reports by natural language processing rather than assigned by a reader looking at the image. They therefore carry an error rate of their own, unknown and not measurable from the data at hand. The condition called *clean* in ?? is the published annotation, not ground truth, so the 5 percent corruption of Task 3.2 is an increment on top of a baseline error whose size is not known. Any true degradation curve is shifted along an axis whose origin is not observed, which compresses the effect that the experiment can see.

One corruption model, one rate, one direction.

The simulation flips a fixed fraction of the positive training labels to negative, independently of the image content. Real annotation error is not independent of the image: a faint effusion is missed more often than a large one, which makes the noise instance dependent and concentrated near the decision boundary, exactly where it is most harmful. The experiment also uses a single rate and a single target finding. Results obtained under class-conditional noise at 5 percent do not transfer by themselves to instance-dependent noise, to higher rates, or to the other thirteen findings. Finally, ChestMNIST has no explicit uncertainty category, unlike CheXpert [3], whose reports distinguish an uncertain mention from a negative one; the uncertainty studied here is simulated for that reason, which is also the gap that Phase IV addresses with a model that represents it directly.

Two smaller points belong in the same list. Model selection uses a single validation split rather than cross-validation, so the selection itself carries a variance that is not quantified. And the posteriors are used only for ranking: no calibration measure is reported, although section 2.1 shows at least one model, Gaussian naive Bayes, whose posterior is badly enough calibrated to

change which metric ranks it first.

2.5 From constraining a model to representing evidence

The robustness mechanism tested in this phase is the only one classical regularisation offers: shrink the hypothesis class so that a handful of wrong labels cannot move the boundary far. It acts on the whole function, uniformly, before any particular image is seen. It gives the model no way of behaving differently on an image that resembles its training data and on one that does not.

That limitation is visible in the outputs these models produce. A sigmoid posterior of 0.5 is returned both for a radiograph that genuinely lies between the two classes, where the evidence is balanced, and for a radiograph unlike anything in the training set, where there is no evidence at all. The two situations are clinically opposite and numerically identical, and no choice of λ , of C or of tree depth separates them, because a point estimate of a probability has no room to carry that distinction. The operating point analysis of section 2.2 makes the same point from another side: the number the model emits is a score to be ranked, and the meaning attached to its value is supplied from outside, by a threshold chosen on validation.

This is the opening that evidential deep learning fills. Rather than emitting a probability, such a model emits the parameters of a Dirichlet distribution over the class probabilities, so that the amount of evidence accumulated for an image and the belief left uncommitted become explicit outputs alongside the prediction [7]. The empirical ground established here, that constraining capacity leaves the degradation under label noise of the same order as the seed-to-seed variability, is what motivates looking for robustness in the output representation rather than only in the size of the hypothesis class. The formal treatment of that argument belongs to Phase IV.

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